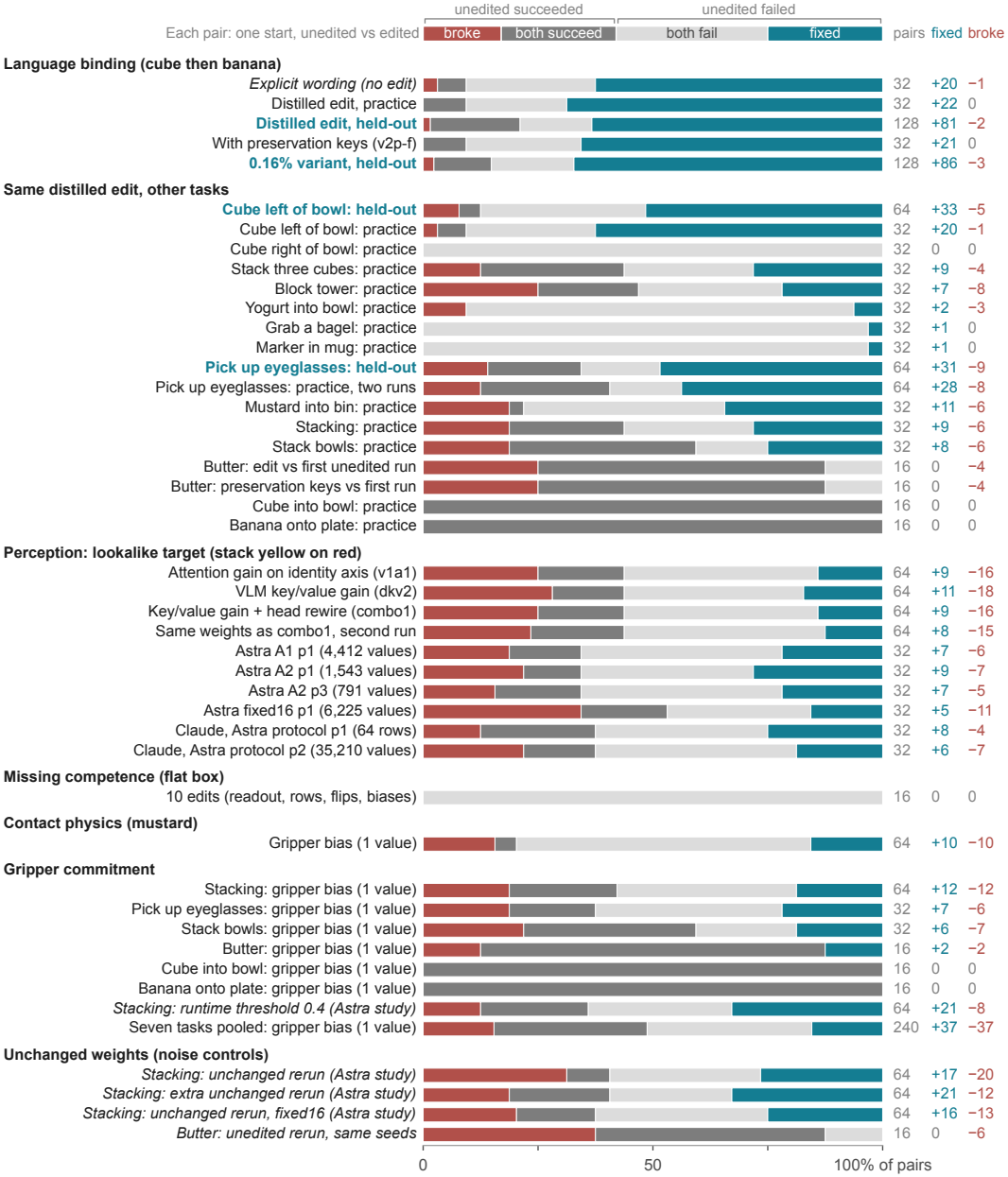

unedited succeeded
unedited failed
Each pair: one start, unedited vs edited
broke
both succeed
both fail
fixed
pairs
fixed
broke
Language binding (cube then banana)
Explicit wording (no edit)
32
+20
−1
Distilled edit, practice
32
+22
0
Distilled edit, held-out
128
+81
−2
With preservation keys (v2p-f)
32
+21
0
0.16% variant, held-out
128
+86
−3
Same distilled edit, other tasks
Cube left of bowl: held-out
64
+33
−5
Cube left of bowl: practice
32
+20
−1
Cube right of bowl: practice
32
0
0
Stack three cubes: practice
32
+9
−4
Block tower: practice
32
+7
−8
Yogurt into bowl: practice
32
+2
−3
Grab a bagel: practice
32
+1
0
Marker in mug: practice
32
+1
0
Pick up eyeglasses: held-out
64
+31
−9
Pick up eyeglasses: practice, two runs
64
+28
−8
Mustard into bin: practice
32
+11
−6
Stacking: practice
32
+9
−6
Stack bowls: practice
32
+8
−6
Butter: edit vs first unedited run
16
0
−4
Butter: preservation keys vs first run
16
0
−4
Cube into bowl: practice
16
0
0
Banana onto plate: practice
16
0
0
Perception: lookalike target (stack yellow on red)
Attention gain on identity axis (v1a1)
64
+9
−16
VLM key/value gain (dkv2)
64
+11
−18
Key/value gain + head rewire (combo1)
64
+9
−16
Same weights as combo1, second run
64
+8
−15
Astra A1 p1 (4,412 values)
32
+7
−6
Astra A2 p1 (1,543 values)
32
+9
−7
Astra A2 p3 (791 values)
32
+7
−5
Astra fixed16 p1 (6,225 values)
32
+5
−11
Claude, Astra protocol p1 (64 rows)
32
+8
−4
Claude, Astra protocol p2 (35,210 values)
32
+6
−7
Missing competence (flat box)
10 edits (readout, rows, flips, biases)
16
0
0
Contact physics (mustard)
Gripper bias (1 value)
64
+10
−10
Gripper commitment
Stacking: gripper bias (1 value)
64
+12
−12
Pick up eyeglasses: gripper bias (1 value)
32
+7
−6
Stack bowls: gripper bias (1 value)
32
+6
−7
Butter: gripper bias (1 value)
16
+2
−2
Cube into bowl: gripper bias (1 value)
16
0
0
Banana onto plate: gripper bias (1 value)
16
0
0
Stacking: runtime threshold 0.4 (Astra study)
64
+21
−8
Seven tasks pooled: gripper bias (1 value)
240
+37
−37
Unchanged weights (noise controls)
Stacking: unchanged rerun (Astra study)
64
+17
−20
Stacking: extra unchanged rerun (Astra study)
64
+21
−12
Stacking: unchanged rerun, fixed16 (Astra study)
64
+16
−13
Butter: unedited rerun, same seeds
16
0
−6
0
50
100% of pairs